

AI ASSISTED CODING LAB – 1

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Batch:24

Lab Assignment -1: Comparing Three Ai Codes

Prompt: Write a python code of prime numbers from 1 to 20

Github Copilot:

```
primes = []  
for num in range(2, 21):  
    is_prime = True  
    for i in range(2, int(num ** 0.5) + 1):  
        if num % i == 0:  
            is_prime = False  
            break  
    if is_prime:  
        primes.append(num)  
  
print("Prime numbers from 1 to 20:")  
print(primes)
```

#output

Prime numbers from 1 to 20:

[2, 3, 5, 7, 11, 13, 17, 19]

Goggle Collab

#code

```
def is_prime(num):  
    if num <= 1:  
        return False  
    for i in range(2, int(num**0.5) + 1):  
        if num % i == 0:  
            return False  
    return True
```

```
prime_numbers = [num for num in range(1, 21) if  
is_prime(num)]  
print(prime_numbers)
```

#output

[2, 3, 5, 7, 11, 13, 17, 19]

Claude

#code

```
def is_prime(n):  
    if n < 2:  
        return False  
    for i in range(2, int(n**0.5) + 1):  
        if n % i == 0:  
            return False  
    return True  
  
primes = [n for n in range(1, 21) if is_prime(n)]  
print(primes)
```

#Output

```
[2, 3, 5, 7, 11, 13, 17, 19]
```

Conclusion:

The three AI-generated Python codes successfully identify and display the prime numbers from 1 to 20. GitHub Copilot, Google Colab, and Claude used slightly different coding approaches, but all produced the same correct output: [2, 3, 5, 7, 11, 13, 17, 19]. This comparison shows that AI tools can generate accurate

and efficient Python programs, while also demonstrating that different AI tools may use different programming styles to solve the same problem.